



E-Farm

Autonomous Greenhouse Harvesting, Engineered Openly

Technical Overview & Vision

An open-source robotics platform for multi-crop greenhouse harvesting

2026 - v1.0

Founded 2022 - Gebze, Bilisim Vadisi (Informatics Valley), Turkiye

Open-source monorepo: github.com/meviza/E-Farm-Autonomous-Harvesting-Robot

This document contains forward-looking targets, explicitly labelled. Figures cited are from the listed public sources.

1. Executive Summary

Greenhouse horticulture across Europe and the Mediterranean is being squeezed by a structural shortage of skilled manual labour. Harvesting is the single most labour-intensive task in protected cropping, and the available workforce is shrinking and ageing. E-Farm is our answer: an autonomous harvesting robot that combines computer vision, LiDAR and RGB-D depth sensing, and a six-axis manipulator with an adaptive gripper to identify ripe produce and harvest it without damaging the plant.

E-Farm was founded in 2022 at Bilisim Vadisi (Informatics Valley) in Gebze, Türkiye. We are a five-person team that has spent roughly three years in research and development, building on open-source hardware and publishing our work openly. We are honest about where we are: this is a credible, working engineering effort moving from prototype toward a pilot and a crowdfunding-backed first production batch - not a finished mass-market product. The figures we present that look ahead are marked as targets.

What E-Farm is

Vision-guided ripeness detection + LiDAR/RGB-D localisation + a 6-axis arm and adaptive gripper, coordinated by an open ROS2/MQTT software stack. Designed for multi-crop greenhouses, built on open-source foundations, and documented in a public monorepo.

2. The Problem

The agricultural labour base is contracting sharply. Across the European Union the farm workforce fell by roughly 3.5 million full-time-equivalent (FTE) workers between 2009 and 2024, and the sector is ageing fast - only about 11% of EU farmers are under 40. Protected horticulture is especially exposed, because harvesting cannot easily be mechanised with conventional field equipment: each fruit must be located, assessed for ripeness, and picked individually without bruising the produce or damaging the plant.

The greenhouse sector this addresses is large:

- Türkiye: approximately 77,600 ha of greenhouse area.
- Antalya region (Türkiye) alone: approximately 31,600 ha.
- Spain: approximately 76,600 ha of greenhouse area.
- Netherlands: approximately 10,260 ha of (high-intensity) glasshouse area.

These are dense, high-value production environments where a reliable autonomous harvester directly addresses the labour gap - and where the structured, repeatable layout of a greenhouse makes robotic navigation and manipulation tractable.

3. Technology

3.1 Computer vision - ripeness detection

Perception is built on a YOLOv8 object-detection model trained through Roboflow on a custom dataset of over 10,000 annotated images spanning five crops. On our project test set the model reaches 98% ripeness-detection accuracy. The model distinguishes ripe from unripe produce and localises each fruit in the image, providing the targets the manipulator acts on.

3.2 Depth sensing and localisation

RGB-D cameras and a LiDAR unit fuse colour, texture and metric depth so the robot can resolve a fruit's position in 3D. The localisation target is sub-centimetre accuracy at the gripper, which is what makes collision-free approach and a clean pick possible inside dense canopy.

3.3 Manipulation - 6-axis arm and gripper

The manipulator is a six-axis arm derived from the open-source PAROL6 design (licensed GPLv3), giving us a fully documented, repairable and modifiable platform rather than a closed black box. It carries an adaptive gripper that conforms to differently sized and shaped fruit and applies controlled force to avoid bruising.

3.4 Power and electronics

The robot runs on a LiFePO4 battery pack managed by an ESP32-based battery management system (BMS). LiFePO4 chemistry is chosen for its safety, long cycle life and stable behaviour in the warm, humid greenhouse environment.

3.5 Software and fleet stack

The control stack is built on ROS2 with MQTT for lightweight telemetry and fleet coordination, allowing multiple robots to be supervised from a single dashboard and enabling future fleet operation.

4. System Architecture

E-Farm follows a clear sense-decide-act pipeline. The perception module (vision + depth) detects and localises ripe fruit. A harvest finite-state machine (FSM) plans the sequence - approach, grasp, detach, place - and dispatches motion commands to the arm controller. Throughout, a telemetry layer streams status, health and yield data over MQTT to the operator dashboard and fleet manager.

The end-to-end flow:

- Perception: YOLOv8 detection + RGB-D/LiDAR depth fusion -> 3D fruit targets.
- Harvest FSM: target selection, motion planning, grasp and detach logic, error recovery.
- Arm: 6-axis PAROL6-based manipulator + adaptive gripper executes the pick.
- Telemetry: MQTT status, yield counters and health metrics to the fleet dashboard.

All of this is developed in the open. The full hardware designs, firmware and software live in our public monorepo at github.com/meviza/E-Farm-Autonomous-Harvesting-Robot, so backers, integrators and researchers can inspect, audit and build on the system.

5. Open-Source Philosophy & Team

E-Farm is built openly on purpose. Agricultural robotics has long been dominated by closed, single-crop systems sold at prices that put them out of reach of most growers, and that cannot be repaired or adapted by the people who use them. We take the opposite stance: by building on documented open-source foundations - the PAROL6 manipulator (GPLv3) and a NASA JPL Open Source Rover-derived chassis - and publishing our own hardware, firmware and software in a single public monorepo, we give growers, integrators and researchers a platform they can understand, trust, repair and extend.

We believe this is also the fastest credible path to reliability. Open development invites scrutiny, reuse of mature components, and contributions from a community that is far larger than any single company's engineering team. It lowers the total cost of ownership for growers and avoids the vendor lock-in that has slowed adoption of agricultural robotics to date.

5.1 Team and approach

E-Farm was founded in 2022 at Bilisim Vadisi (Informatics Valley) in Gebze, Türkiye - one of the country's principal technology hubs. The core team of five spans computer vision, robotics and mechatronics, embedded systems and power electronics, and software/fleet infrastructure. Over roughly three years of R&D we have moved from individual subsystems to an integrated prototype: a trained perception model, a working manipulator and gripper, a LiFePO4 power system with a custom BMS, and a ROS2/MQTT software stack

tying them together.

Our working method is deliberately incremental and evidence-driven. We validate each subsystem against measurable targets - detection accuracy, localisation error, pick success - before integrating it, and we keep the whole system in the open so progress can be verified rather than merely claimed.

6. Deployment & Economics

E-Farm is designed for the structured environment of a commercial greenhouse: produce grown in rows on predictable supports, with level or gently uneven floors and controlled lighting. This structure is what makes autonomous harvesting tractable - navigation follows known rows, and the manipulator works within a bounded, repeatable geometry.

A typical deployment proceeds in stages:

- Survey and mapping of the greenhouse rows and crop layout.
- Per-crop perception tuning using the existing trained models as a base, refined on the grower's produce.
- Supervised pilot operation alongside staff, building confidence and capturing edge cases.
- Routine autonomous harvesting with remote monitoring via the MQTT fleet dashboard, and eventual multi-robot operation.

The economic case rests on the labour shortage described in Section 2. Where skilled harvesting labour is scarce, expensive or seasonal, a robot that works long shifts, reports its yield in real time and scales as a fleet addresses both cost and availability. Concrete throughput and cost-per-kg figures depend on crop, layout and configuration, and will be reported transparently from pilot data rather than promised in advance.

7. Market

The agricultural-robots market is growing quickly. According to MarketsandMarkets (2025), the global agricultural-robots market is valued at about USD 17.73 billion in 2025 and is projected to reach USD 56.26 billion by 2030 - a compound annual growth rate (CAGR) of roughly 26%. Harvesting robotics is one of the fastest-growing segments within that market, driven directly by the labour shortage described above.

Market at a glance (MarketsandMarkets, 2025)

2025: USD 17.73B -> 2030: USD 56.26B | CAGR ~26%

8. Competitive Context

Robotic harvesting is an active field. The notable players each tend to focus on a single crop or geography:

Company / Platform	Focus
Octinion / Rubion (BE)	Strawberry picking; table-top systems.
Agrobot (ES)	Strawberry harvester with multi-arm gantry.
Root AI / Virgo (US)	Greenhouse tomato/cucumber harvesting.
Tortuga AgTech (US)	Strawberry/grape robotic harvest-as-a-service.
FFRobotics (IL)	Multi-gripper orchard fruit harvester.
Four Growers (US)	Greenhouse tomato harvesting robot.
Saga Robotics / Thorvald (NO)	Modular agri-robot platform, UV + harvest.

E-Farm's positioning is distinct on three axes. First, geography and cost base: we are built in Türkiye, close to

one of the world's largest greenhouse footprints, with a competitive engineering cost structure. Second, openness: the platform is open-source end-to-end, which lowers the barrier for integrators and researchers and avoids vendor lock-in. Third, scope: we target multi-crop greenhouse harvesting rather than a single fruit, using a general-purpose vision + 6-axis manipulation approach.

9. Risks and Roadmap

Honest risk note

E-Farm is a small team productising an R&D prototype. Real risks remain: field reliability and pick success rates in dense canopy, cycle-time and throughput economics versus manual labour, manufacturing and supply-chain execution for a first hardware batch, and regulatory/safety certification. Forward-looking dates and performance figures in this document are targets, not guarantees.

Roadmap (dates beyond the present are targets):

- 2025 - Pilot: field trials of the integrated prototype in working greenhouses. (target)
- 2026 - Crowdfunding campaign and first production batch. (current)
- 2027 - Fleet: multi-robot deployments coordinated through the ROS2/MQTT stack. (target)

10. Selected References

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